

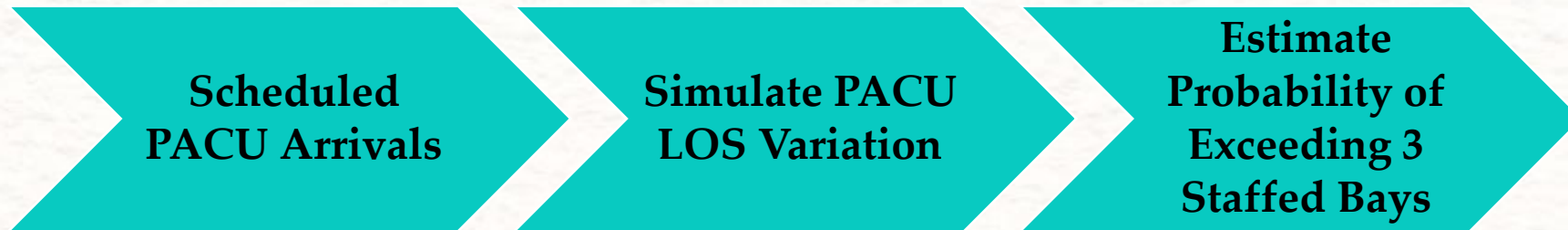
PACU Capacity Stress Simulation For Capacity Planning

Simulation-based staffing trigger for a short-staffed 3-bay PACU

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Decision Question: When does a 3-bay PACU need help?

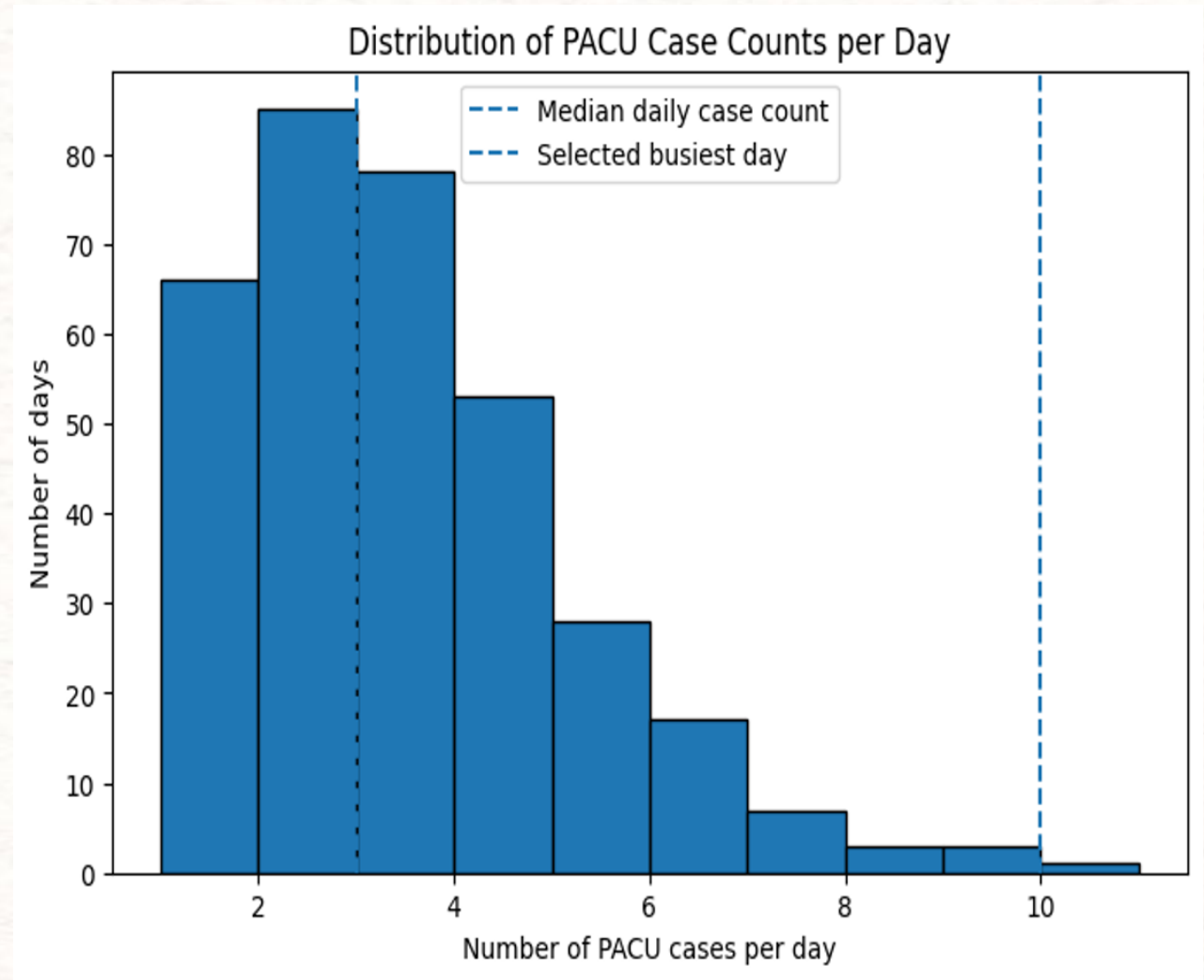
Use the morning schedule to estimate whether PACU demand is likely to exceed **3 staffed bays** before the day becomes overloaded.



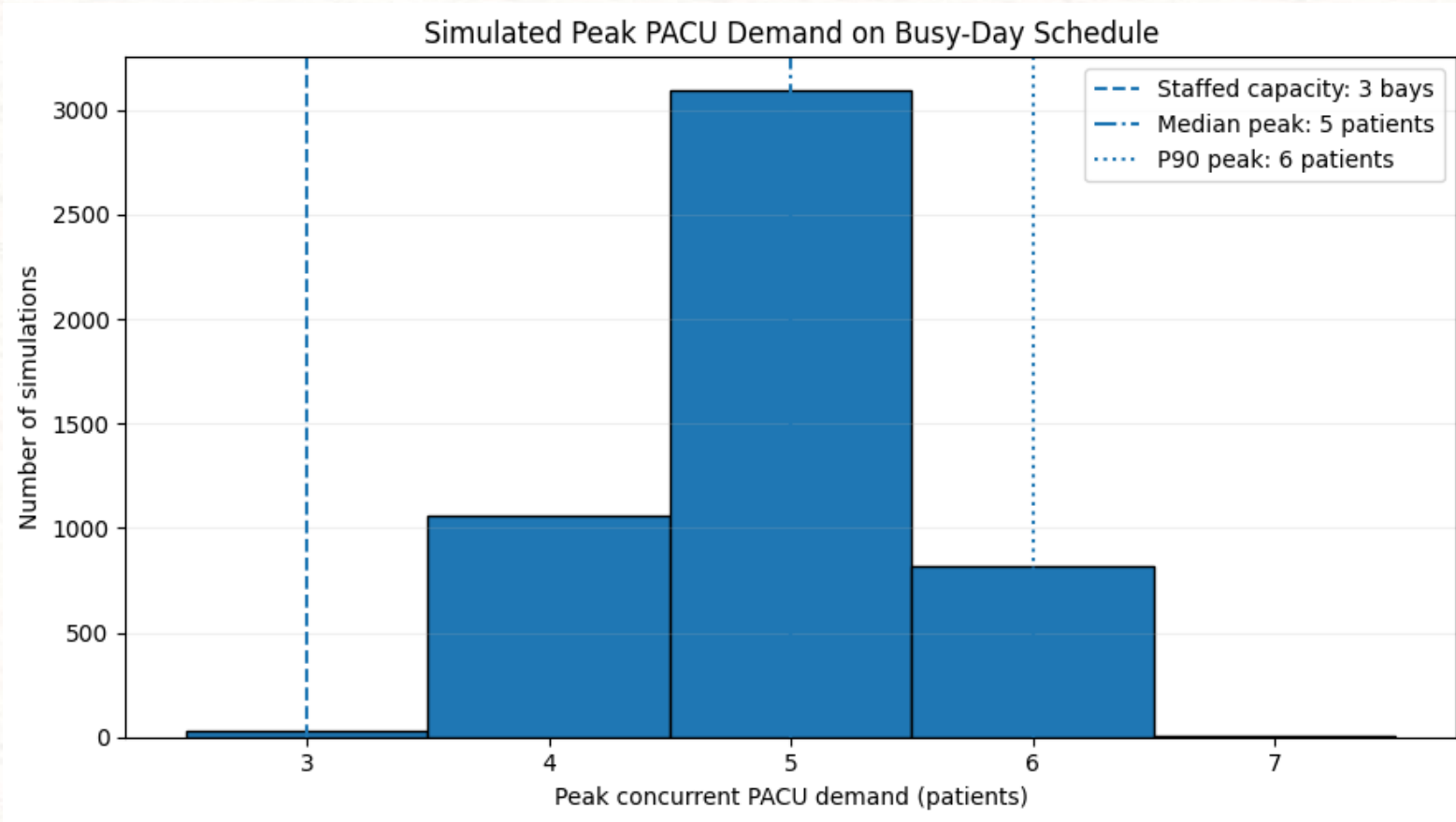
Stress-Test Schedule: 10 PACU Arrivals

- **March 30, 2024**
 - Highest-Volume Observed Day
 - 10 PACU arrivals
 - Median day = 3 arrivals

Used as a high-volume schedule to test 3-bay staffing resources.



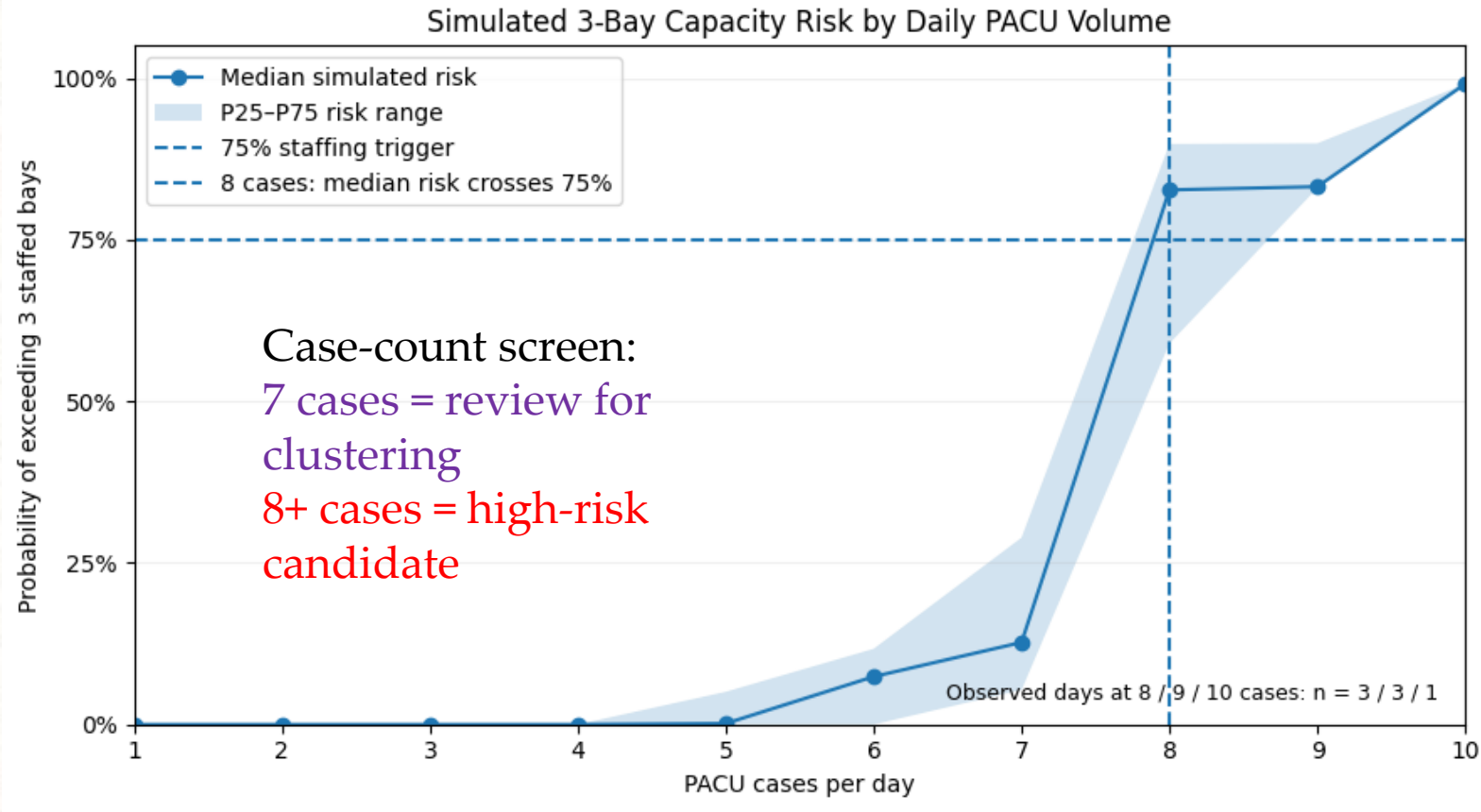
The 3-bay schedule exceeds capacity in 99.4% of simulations



- **99.4%:** Chance of exceeding 3 bays
- **79.6 minutes:** Expected time above capacity
- **5 patients:** Median simulated peak
- **6 Patients:** P90 simulated peak

Demand is likely to exceed capacity. The typical peak is 2 patients above staffed capacity.

Simulated 3-Bay Over-Capacity Risk by Daily PACU Case Count

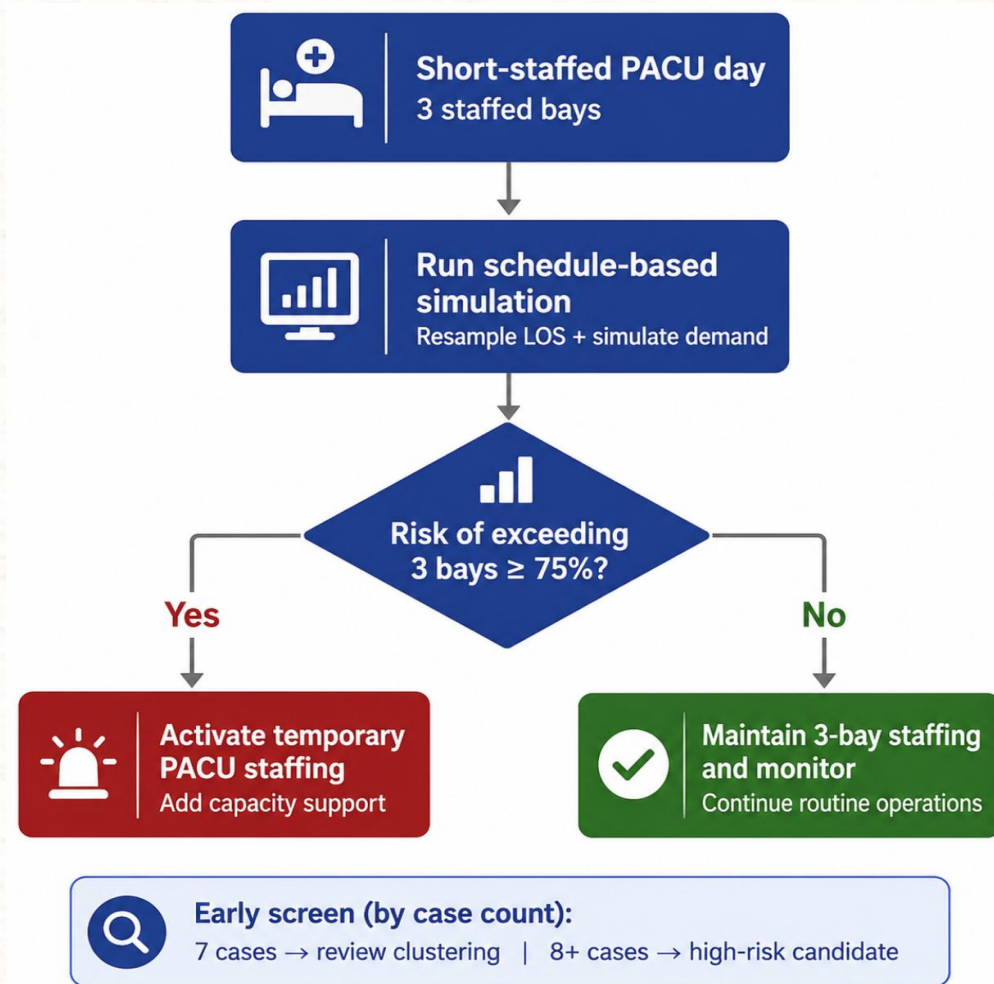


Recommended rule:

- Run the simulation before / early in the day.
- $\geq 75\%$ Simulated Risk: Activate Temporary Staffing

Case count helps flag risk, but the final decision should come from the simulated over-capacity probability.

Operational Staffing Rule



The rule gives leaders a clear morning decision:
Staff proactively when the schedule is likely to
exceed 3 staffed bays.

Example: Stress-test day result: 99.4% risk →
Activate support